

According to Timothy Lu, associate professor of electrical engineering and computer science, and biological engineering at MIT, “By combining engineered biological sensors together with low-power wireless electronics, we can detect biological signals in the body and in near real time, enabling new diagnostic capabilities for human health applications.”



To make bacteria more useful for real-world applications, MIT researchers combined them with an electronic chip that could translate the bacterial response into a wireless signal (Credit: MIT)

The researchers created sensors that respond to haem, a component of blood, and showed that these work in pigs. They also designed sensors that can respond to a molecule that is a marker of inflammation.

Fabric that can change colour on demand

A team of scientists at CREOL, the College of Optics and Photonics at University of Central Florida (UCF), has developed the first-ever, active user-controlled colour-changing fabric. The fabric allows the user/wearer to change the colour or pattern of the fabric through a smartphone.

The colour-changing fabric is different from previous colour-changing ones in the market, which contain LEDs that emit



A solid purple tote bag can gradually add blue stripes when the user presses Stripe on the app, or switch to a solid blue (Credit: www.engadget.com)

light of various colours. CREOL's ChroMorphous technology enables a user-controlled dynamic colour and pattern change in large woven fabrics and cut-and-sewn textile products. Each thread woven into the fabric incorporates within it a thin metal micro-wire. Electric current flows through the micro-wires, slightly raising the thread's temperature. Special pigments embedded in the thread then respond to this modification of temperature by changing its colour.

The technology allows the user to control when the colour change happens and what pattern to appear on the fabric. Using an app, the wearer can choose from a variety of patterns and colours to suit his or her need, or mood, all from a smart device.

Wearable chair that lets you sit anywhere

A portable contraption that looks nothing like a chair, Chairless Chair allows users to sit whenever and wherever they want. This flexible, ergonomic device looks more like an exoskeleton and extends from the hip to the backs of the feet, making them look a little bit like humanoid cyborgs, at least from the back and waist down. All it takes to work in a healthy position is putting on the device.



Chairless Chair allows for quick, easy and flexible changes between sitting, standing and walking (Credit: www.noonee.com)

Created by noonee AG, a German startup, Chairless Chair is made using hi-tech composite polymers. It weighs just 3.5 kilograms and adapts to different body sizes and safety shoes. According to the chair's inventor, Keith Gunura, the main application is for manufacturing companies, where workers are required to stand for a long period of time and sometimes bend into unnatural positions to assemble a product.